Computer Setup Guide

for Data Analytics Course (Undergraduate)

R, RStudio, and course materials

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Contents

This guide walks you through installing everything you need for this course: Git, R, RStudio, a LaTeX distribution, and TeXStudio. It covers both Windows and Mac — follow only the steps for your operating system. By the end, you will have the course materials downloaded to your computer and be ready to run R and LaTeX.

You do not need a GitHub account for this course. The course repository is public, so you can download it directly — no sign-up required.

1 Software Installation

- Git — the tool used to download (and later update) the course materials.
- R — the programming language used in this course.
- RStudio — the application you will actually work in; it uses R behind the scenes.
- A LaTeX distribution — the underlying software that turns LaTeX documents into PDFs.
- TeXStudio — the editor you will use to write and compile LaTeX documents.
- MS Excel — used occasionally alongside R for basic spreadsheet tasks.

Install them in this order: Git, then R, then RStudio, then your LaTeX distribution, then TeXStudio. This takes about 30–40 minutes total (the LaTeX distribution is a large download).

Git, R, RStudio, your LaTeX distribution, and TeXStudio are all free and open-source. Microsoft Excel is the only paid tool used in this course — see the note below.

1.1 Git

Windows

1. Go to `https://git-scm.com/download/win` — the download should start automatically.
2. Run the installer. You can accept all the default options by repeatedly clicking "Next," then "Install."
3. Once finished, open the Start menu, search for "Git Bash," and open it.
4. Type the following and press Enter:

   ```
   git --version
   ```

If you see a version number (e.g. `git version 2.44.0`), Git is installed correctly.

Mac

1. Open the Terminal app (press Cmd+Space, type "Terminal," press Enter).
2. Type the following and press Enter:

   ```
   git --version
   ```

3. If Git is not installed yet, macOS will show a popup asking to install the "command line developer tools." Click Install and wait a few minutes for it to finish.
4. Once it is done, run `git --version` again to confirm you see a version number.

> **Note:** If no popup appears and you get a "command not found" error instead, download Git directly from `https://git-scm.com/download/mac` and run the installer.

1.2 R

1. Go to `https://cran.r-project.org`.
2. Windows: click "Download R for Windows" → "base" → the top download link (e.g. "Download R-4.x.x for Windows").
3. Mac: click "Download R for macOS" and choose the correct file for your Mac — the page lists an "arm64" build (for Apple Silicon Macs: M1/M2/M3/M4) and an "x86_64" build (for older Intel Macs). If you are not sure which chip your Mac has, check Apple menu → About This Mac.
4. Run the installer and accept all default options.

> **Note:** You will not open R directly in this course — you will always work through RStudio, which uses R automatically in the background.

1.3 RStudio

1. Go to `https://posit.co/download/rstudio-desktop/`.
2. Scroll to "Install RStudio Desktop" and download the free version for your operating system.
3. Run the installer and accept all default options.
4. Open RStudio once to confirm it launches correctly. You should see a window with several panels (Console, Environment, Files, etc.).

1.4 LaTeX distribution

Windows

1. Go to `https://miktex.org/download`.
2. Download and run the Basic MiKTeX Installer, accepting the default options.
3. The first time you compile a document, MiKTeX may prompt you to install missing packages automatically — choose "Install" when this happens.

Mac

1. Go to `https://tug.org/mactex/`.
2. Download MacTeX and run the installer, accepting the default options.

> **Note:** MacTeX is a large download (several GB) and can take a while depending on your internet connection — start this install early if possible.

1.5 TeXStudio

1. Go to `https://www.texstudio.org/`.
2. Download the installer for your operating system and run it, accepting the default options.
3. Open TeXStudio once to confirm it launches correctly.

1.6 Microsoft Excel

Microsoft Excel is used in this course alongside R for basic spreadsheet tasks. Unlike the other tools in this guide, it is a paid product, so installation steps are not covered here.

> **Note:** If you do not already have Excel through a personal Microsoft 365 subscription, check with your university — many institutions provide free or discounted Microsoft 365 licenses (including Excel) to enrolled students.

2 Download the course materials

The course repository is public, so you can download it without creating any account. You will use Git to "clone" it — this downloads the whole folder structure (scripts, data, slides) to your computer.

1. Choose (or create) a folder on your computer where you want the course materials to live. For example, create a new folder called `DataAnalytics` on your Desktop. (Avoid spaces in the folder name — they cause problems in the next step.)
2. Get the full path to that folder:
   - **Mac:** in Finder, hold the `Option` key and right-click (or Control-click) the folder, then choose "Copy '`DataAnalytics`' as Pathname."
   - **Windows:** in File Explorer, hold `Shift` and right-click the folder, then choose "Copy as path."
3. Open Terminal (Mac) or Git Bash (Windows). Type `cd ` (with a space after it), then paste the path you just copied, and press Enter. For example, on Mac:

   ```
   cd /Users/yourname/Desktop/DataAnalytics
   ```

   On Windows, "Copy as path" gives a `C:\...` style path wrapped in quotes — paste it as-is, quotes included:

   ```
   cd "C:\Users\yourname\Desktop\DataAnalytics"
   ```

4. Clone the repository:

   ```
   git clone https://github.com/vojtechzika/data-analytics-ug.git
   ```

5. This creates a new folder called `data-analytics-ug` inside the folder you selected, containing all the course files.

6. Throughout the course, new materials may be added to the repository. To get the latest version at any point, open Terminal or Git Bash and `cd` into the `data-analytics-ug` folder itself (one level deeper than before), then run:

   ```
   cd /Users/yourname/Desktop/DataAnalytics/data-analytics-ug
   git pull
   ```

   This downloads any new or updated files without affecting your own work, as long as you have not edited the same files the instructor changed.

3 Open the project in RStudio

1. Open the `data-analytics-ug` folder you just downloaded, in Finder (Mac) or File Explorer (Windows).

2. Double-click `data-analytics-ug.Rproj`.

3. RStudio should open with this project loaded — you will see "data-analytics-ug" in the top-right corner of the RStudio window.

4. In RStudio, click into the Console panel (usually bottom-left).

5. Type the following and press Enter:

   ```
   1 + 1
   ```

6. You should see `[1] 2` printed below.

7. In the Files panel (bottom-right), confirm you can see the `R`, `data`, and `slides` folders.

If all of the above worked, your setup is complete.

4 Troubleshooting

- **"git is not recognized" / "command not found"** right after installing: close and reopen Terminal or Git Bash (or restart your computer), then try again.

- **Nothing happens when double-clicking the `.Rproj` file:** make sure RStudio is installed, then right-click the file and choose "Open with → RStudio."

- **`git clone` fails with a permissions or authentication error:** double-check the URL was typed exactly as shown — since the repo is public, no login should be requested.